Ethics, limitations, and course recap

Lecture 12 – Where does deep learning still break? And what to do about it

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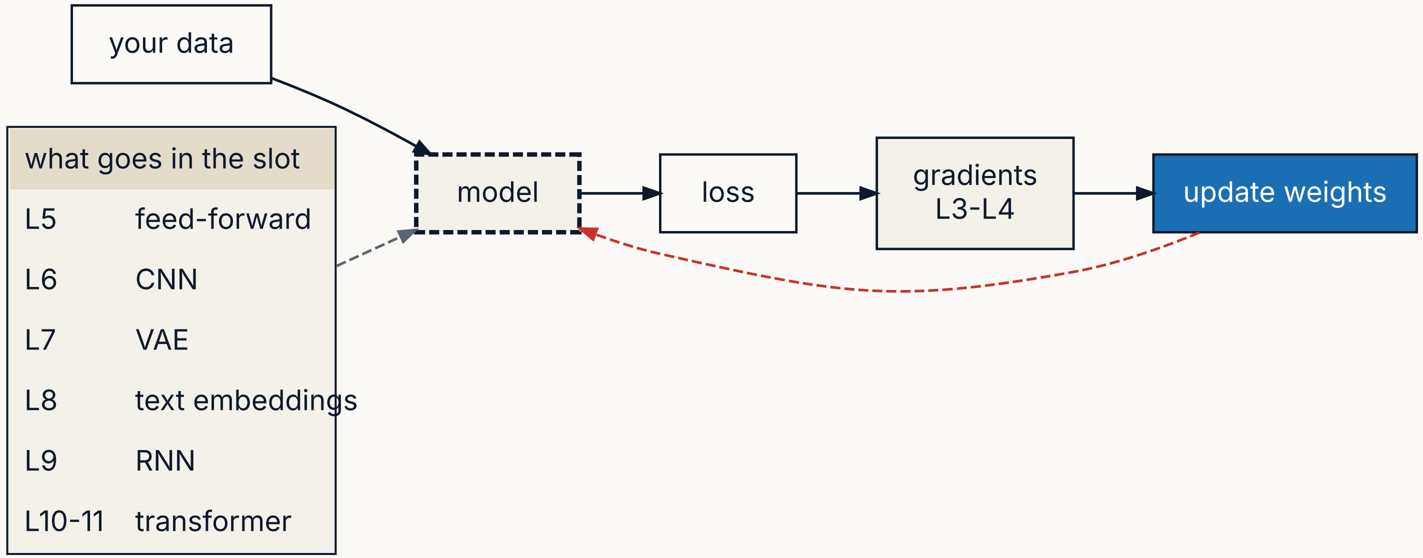
LSE SUMMER SCHOOL · 2026

The final lecture!

How does it all fit together?

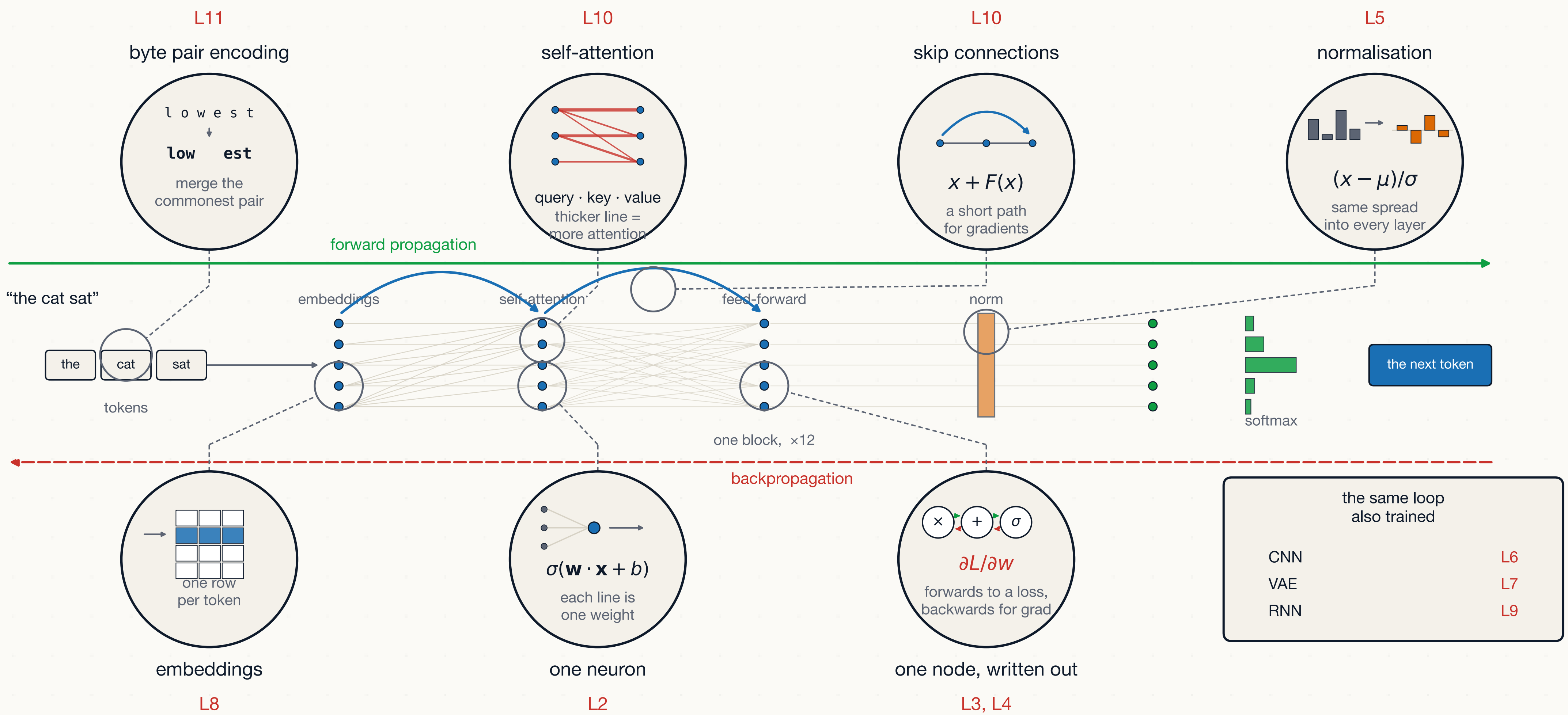
TWELVE LECTURES, ONE DIAGRAM



- **L1-L4 provides the compute engine.** A computational graph + the chain rule to propagate loss gradients.
- **L5 built up a full loop.** I.e. the red arrow above — forward pass, backprop, repeat. We haven't changed that loop since.
- **L6-L11 changed only the configuration of nodes.** Each architecture is a different **model**.

Where does each lecture show up?

WE'VE BUILT ALL OF THIS!



Five ideas that connect everything

1. **Composition.** A network is a sequence of cheap operations (matmul + non-linearity). Stacking gives expressiveness.
2. **Differentiability.** Every operation has a known gradient. The chain rule glues them.
3. **Autodiff.** Backpropagation = reverse-mode autodiff, which has a linear cost in graph size.
4. **Optimisation.** Define a loss. Step against the gradient. Repeat.
5. **Scale.** Apply the same building blocks with 10^{12} parameters and you can express more.

What's *not* in this course

A LIST OF THINGS WORTH KNOWING THAT WE DIDN'T COVER

- Reinforcement learning — RLHF (reinforcement learning from human feedback) and its optimisers, PPO (proximal policy optimisation) and DPO (direct preference optimisation)
- Graph neural networks
- Diffusion models in depth (we sketched)
- Self-supervised pre-training methods
- Multi-modal models (text + image jointly)
- Long-context architectures (Mamba, RingAttention)
- Distillation, model compression
- Federated learning
- Causal ML and double machine learning
- Bayesian deep learning
- MoE (mixture of experts) implementation
- Speculative decoding

Each of these could be a lecture/course in itself, but you have the foundations to read into any of them now.

Part 2 · Limitations + ethics

Where deep learning is *bad* — even at scale

Data-poor regimes

- Strong supervised performance needs thousands to millions of labelled examples.
- Many small-data social-science problems work better with “classical” models.

Long-horizon reasoning

- LLMs handle short, local reasoning well.
- Long multi-step proofs, planning, mathematical rigour — still brittle.
- Chain-of-thought helps. It doesn’t solve.

Where deep learning *can't* do at all

- **Causal inference from observational data.** Still needs assumptions you encode by hand — DL doesn't substitute for instrumental variables, RCTs, etc.
- **Out-of-distribution generalisation.** Same as any ML model, it will break when the test data diverges from training.
- **Calibrated uncertainty.** "Knowing what they don't know" is hard (we're seeing this at the moment)
- **Anything resembling agency or intent.** A trained model has no goals, no beliefs, no plans.

Anthropomorphising models is dangerous.

⚠ **LLMs hallucinate by design.**

They produce *plausible continuations*, not *true statements*.

Environmental impact

BEYOND "AI USES A LOT OF COMPUTE"

Training

- GPT-3 training: estimated **552 tonnes CO_2** (one-time cost).
- GPT-4: estimated 10×–100× more.
- Frontier models now train multiple times per year.

Inference

- The "folk number" says one ChatGPT query costs ~10× a Google search — but *treat this with caution*
- Aggregated across hundreds of millions of users per day, **inference dominates** training in cumulative impact.
- Data centre water use for cooling is a separately growing concern
  - And datacentres more generally are an increasingly salient political issue

One query costs little

The footprint of these systems stems from their use *at scale* and depends on **where** data centres are sited.

- Moving to renewables/non-fossil fuels helps, but this is a policy question we are still figuring out.

Where bias comes from

Some (but not all) of the sources of bias:

1. **Data sampling** — what shows up in the training set
2. **Label noise** — when humans encode their own biases as ground truth
3. **Construct mismatch** — when the modelled variable is a proxy for what we actually care about (e.g. “creditworthiness” vs “ability to pay”).
4. **Optimisation target** — minimising aggregate error can smooth over important subgroup differences
5. **Deployment context** — a model trained on one population deployed on another.
6. **Feedback loops** — biased predictions become future training data.

Technical fixes only address (1) and (4).

The rest are governance, methodology, and political problems – that’s why these companies like hiring social scientists!

Bias

Models reflect their data. Data reflects the world. Both have problems.

Previously discussed examples

- Facial recognition with much higher error rates on dark-skinned faces (Buolamwini & Gebru 2018).
- Hiring screeners that downweight applicants from women's colleges.
- Healthcare algorithms allocating less care to Black patients (Obermeyer et al. 2019).
- LLMs producing racist completions of innocuous prompts.

PATTERNS ARE BIAS

- "It's just learning patterns"
- Benchmarks that measure narrowly can hide disparate impact
- Proxies remain even after projecting out a dimension

Mitigation strategies for bias

WHAT PARTIALLY WORKS

PRE-PROCESSING

Adjust the training data — rebalance subgroups, reweight, or filter.

- Easy to implement.
- Hard to specify "balanced".
- Doesn't address (5) or (6).

IN-PROCESSING

Add fairness constraints to the loss — penalise disparate outcomes during training.

- Many constraint definitions exist (and conflict).
- Trade-offs with accuracy.

POST-PROCESSING

Adjust model outputs after training to equalise outcomes across subgroups.

- Doesn't require retraining.
- Often the most tractable, especially when deploying third-party models.

> **Partial solutions**
>
> Different notions of fairness can *conflict with one another* — you sometimes cannot satisfy them all at once.

Case study · the Dutch childcare benefits scandal

THE *TOESLAGENAFFAIRE* · NETHERLANDS, MID-2010S TO 2021

From the mid-2010s, the Dutch tax authority used a risk-scoring model to flag childcare-benefit claims for fraud investigation.

What happened

- **Tens of thousands of families** were wrongly accused of benefit fraud
- An accusation meant repaying benefits in full with the debt due at once
    - Often **tens of thousands of euros**
- Families were pushed into debt, lost homes, and/or jobs over money they didn't owe
- In **January 2021**, the Dutch government resigned over the scandal

The technology was not "futuristic" in any sense:

- A risk score computed from family features
- Speaks to the various traps and issues we face when building effective DL models

Case study · where it broke in our terms

FOUR FAILURES, FOUR CONCEPTS YOU ALREADY KNOW

1. **The labels were not ground truth.**
   - The model learned from past fraud *investigations*, not convictions.
   - “Flagged before” became “high risk now”
2. **Proxies did the work.**
   - Nationality sat in the feature set
   - The Dutch data-protection authority later ruled its use unlawful
   - Low income also pushed the score up
3. **A high score triggered automatic action**
   - By default, flagged families had benefits stopped, repayment demanded, without any review
   - The score *became* the decision.
4. **The feedback loop never happened.**
   - Wrongly flagged families winning their appeals arrived too late to change it.
   - In training terms, no gradient ever flowed back to keep weights up to date/relevant

Where did the bias come from?

Label noise (2), construct mismatch (3), feedback loops (6).

Case study · what would have helped

- **Audit error rates by group.**
  - Aggregate accuracy can look fine while one group carries the errors.
- **Keep a human who can overrule the score.**
  - Need to have the authority, time, and information to say the model is wrong about *this* family
- **Contestability with deadlines.**
  - Those deadlines have to be set to ensure the model can be updated
- **Check the model against the world.**
  - The training data was wrong about the world!

Pre-, in-, and post-processing corrections all apply here

None of them can tackle institutional or social sources of bias (including how the model may have been validated/signed-off)

Stop & check — can a model be “fair”?

STOP & CHECK · 60 SECONDS

TRY

A recidivism-risk model is **calibrated**: among everyone it scores “high risk”, the same fraction actually reoffend — *regardless of race*. A journalist then shows that, among people who did **not** reoffend, Black defendants were flagged “high risk” far more often than white defendants.

Is the model broken?

THE REVEAL

This example is taken from the **COMPAS** dispute (ProPublica vs. Northpointe, 2016). *Both sides made arithmetically correct claims!*

When the base rate of reoffending differs across groups, **calibration** and **equal false-positive rates** cannot both hold at once (except in trivial cases). You are forced to *choose* which criterion to satisfy (**Chouldechova 2017**)

That choice is **normative** and no amount of engineering makes it go away.

Hallucination + misuse

INHERENT

- LLMs invent citations, statistics, court cases.
- No internal “truth” signal — only plausibility.
- Mitigations (retrieval-augmented generation — RAG — tool use, fine-tuning) reduce but don’t eliminate.

Weaponised

- Phishing at scale.
- Personalised disinformation.
- Synthetic media abuse (deepfakes, non-consensual intimate imagery — NCII).
- Code that *almost works* — subtly broken security.

How would you know if it's any good?

THE QUESTION THIS COURSE HAS BEEN QUIETLY DODGING

For twelve lectures, validation meant one thing: hold out some data, compute the loss on it, compare. That works when each input has one right answer and you have it written down.

Ask a model to summarise a document, draft a reply, or code a variable from an interview transcript. There are many good answers and no key to mark against. The held-out loss still computes — it just stops telling you what you want to know.

Three questions get run together under the word "validation":

1. **Is it capable?** How good is this model, in general?
2. **Is it valid for my task?** Does it measure the thing I say it measures, on my data?
3. **Is it safe to deploy?** What happens when it fails, or when someone attacks it?

Tip

Different questions, different evidence. Most arguments about whether an AI system "works" are two people answering different ones.

Four kinds of evidence

Benchmarks

A fixed set of questions with known answers. MMLU, GPQA, SWE-bench.

- Cheap, repeatable, comparable across models.
- Measures the benchmark, not your task.

Task-specific held-out sets

Your own data, labelled by hand, held back.

- The only evidence about *your* task.
- Costs human hours. Goes stale as the model changes.

Human preference

Show people two outputs, ask which is better. Chatbot Arena does this at scale.

- Captures quality no metric writes down.
- Measures what people *like*. Fluent and wrong beats terse and right.

Deployment monitoring

Watch the system in use — failures, complaints, appeals, subgroup error rates.

- The only evidence about the real distribution.
- Arrives after the harm.

> **Warning**
>
> The Dutch case failed at all four. No held-out audit by group, no monitoring that closed the loop inside a decade.

Why benchmarks stop working

TWO FAILURE MODES, BOTH WELL DOCUMENTED

Saturation

MMLU (Hendrycks et al., 2021) — 15,908 questions across 57 subjects, released 2020.

- GPT-3 (2020): **43.9%**.
- Frontier models by 2024–25: **88–90%**.
- Estimated human-expert ceiling: **89.8%**.

At the ceiling, the benchmark stops separating models. Everyone scores the same.

Contamination

Benchmarks are public, so they end up in the training corpus. The model has seen the test.

Zhang et al. (2024) (Zhang et al., 2024) commissioned **GSM1k** — 1,000 fresh grade-school maths problems in the style of the standard GSM8k benchmark.

- Some model families scored up to **13 points lower** on the fresh set.
- Frontier models dropped much less.

The response: build benchmarks that resist both. GPQA (Rein et al., 2024) is written to be *Google-proof* — PhD-level experts score **65%**, while skilled non-experts with over 30 minutes of unrestricted web search score **34%**. Others keep the test set private, or commission a fresh one each year.

How much does a two-point lead mean?

A BENCHMARK SCORE IS A SAMPLE MEAN. TREAT IT LIKE ONE.

A model answers **400** benchmark questions and gets **90%** right. That proportion has a standard error:

$$\mathrm{SE} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.9 \times 0.1}{400}} = \sqrt{0.000225} = 0.015$$

So 1.5 percentage points, and a 95% interval of roughly **87.1% to 92.9%**.

Now a rival model scores **91%** on its own 400 questions. The standard error of the *difference* is about **2.1 points** — larger than the 2-point lead itself. The leaderboard gap is noise.

Tip

Miller (2024) (Miller, 2024) sets out the fixes: report standard errors, **cluster** them when questions share a passage (up to **3×** the naive figure), and run both models on the *same* questions so you can analyse paired differences — which shrinks the error bar a long way.

Using a model to mark a model

LLM-AS-JUDGE — THE CURRENT WORKHORSE

Human rating is the gold standard and costs pounds per comparison. So: prompt a strong model with the question, the two answers, and a rubric, and ask it which is better.

Does it work?

Zheng et al. (2023) (Zheng et al., 2023) checked GPT-4's judgements against human preferences on MT-Bench.

- Judge agrees with humans: **≈85%**.
- Two humans agree with each other: **81%**.

The judge matches human agreement levels — which is the ceiling, not a licence.

Known biases

- **Position** — which answer was shown first.
- **Verbosity** — longer answers score higher.
- **Self-preference** — a model rates its own output above a rival's.

Mitigations: run each pair in both orders and average, give the judge a reference answer, and calibrate against a human-labelled subset first.

> **Warning**
>
> The judge is a model with the same blind spots as the thing it is judging. It will not catch a mistake both of them make.

Stop & check — is 90% agreement good enough?

STOP & CHECK · 60 SECONDS IN PAIRS

TRY

You use an LLM to label 10,000 news articles for whether they are about immigration. You hand-code 100 yourself to check. The model agrees with you on **90** of them.

Is that good enough to run your regression on the other 9,900?

THE REVEAL

Suppose 10 of your 100 articles really are about immigration, and the model finds 5 of them while flagging 5 articles that aren't.

- Accuracy: $(5 + 85)/100 = \mathbf{90\%}$.
- A model that answers "no" every time: also $\mathbf{90\%}$. Your model has beaten it by nothing.
- It misses **half** the immigration articles, and half of what it flags is wrong.

Cohen's κ rescales agreement against chance: $\kappa = (p_o - p_e)/(1 - p_e)$, with p_o the observed agreement and p_e what two raters would hit at random. Both say yes 10% of the time, so $p_e = (0.1)(0.1) + (0.9)(0.9) = 0.82$:

$$\kappa = \frac{0.90 - 0.82}{1 - 0.82} = \frac{0.08}{0.18} \approx 0.44$$

What to do about the other 9,900

THE STATE OF THE ART FOR LLM LABELS IN RESEARCH

Egami et al. (2023) show how important tackling this bias is for inferential research:

- Feeding LLM labels straight into a downstream regression gives **biased coefficients and confidence intervals**
- The bias comes from the errors being *correlated with the thing you are studying*.

The fix

Keep a random, hand-coded **gold-standard subset**. Run the model on everything, then use the subset to correct the estimate and the standard errors.

- **DSL** (design-based supervised learning) in political science.
- **PPI** (prediction-powered inference), Angelopoulos et al. (2023) in *Science* (Angelopoulos et al., 2023) — same idea, same guarantee.

WHAT YOU GET

- Valid inference **even if the labels are badly biased**.
- Tighter intervals when the model is good, correct ones when it isn't.
- The labour cost is relatively small — a few hundred documents, coded properly.

"Can I use an LLM to code my data?"

Yes, of course! So long as you also hand-code enough of it to find out how wrong the model was.

How do you test for harm?

CAPABILITY EVALUATION

Where accuracy asks *how often is it right*, **safety** evaluation asks *what is the worst thing it will do if someone tries*.

Red-teaming

People — and now models — attack a system to make it fail.

E.g. the UK AI Safety Institute (AISI) tested 5 public LLMs in **May 2024**

- All were highly vulnerable to **basic** jailbreaks
- Some produced harmful outputs with no jailbreak attempt

Dangerous-capability evaluations

Some things we should just always check before releasing:

- Can it assist a cyber-attack, or with chemical and biological weapons?
- Can it act on its own over long horizons?
- Does it “uplift” the capability of non-experts beyond the tools they already have?

Anthropic’s Responsible Scaling Policy, OpenAI’s Preparedness Framework, Google DeepMind’s Frontier Safety Framework are all *internal* efforts to exercise this caution

A failed elicitation is not proof of absence

This attempt didn’t find the capability, but someone with more time, better prompts, or fine-tuning access may. See [this interesting interview]

Privacy and data extraction

TWO RELATED CONCERNS

Training-data privacy

- With so many parameters, LLMs can memorise. Specific email addresses, anecdotes, and API keys can be recovered from production models via prompt attacks.
- Carlini et al. (2021): GPT-2 can be made to emit memorised training data

Inference-time privacy

- Every prompt you send to an API is data the provider can store, train on, or sell.
- OpenAI explicitly logs interactions (with opt-out).
- On-device / open-weights models avoid this entirely — but at a quality cost.

Never send identifiable records to a closed API (without explicit consent and clearance)

Use your university's research-ethics committee (in the US, an Institutional Review Board, or IRB; health data carries extra rules such as the US HIPAA). Most commercial APIs are *not* GDPR-compliant by default.

AI safety and the existential question

As models become more capable, they may (be able to) pursue goals misaligned with human intent (e.g. hacking a yoga booking system)

- **Mechanistic interpretability** — understand what circuits inside a model do.
- **Alignment research** — methods to make models do what we want, not what we incentivise.
- **Evaluations + red-teaming** — test models for dangerous capabilities before release.

> ⚠ **Warning**
>
> **This is all *very* contested.** Some serious researchers think this is the most important problem of the century. Others think it's overblown.
>
> You *should* know the debate exists, and have read at least one careful piece from each camp.

Independent safety organisations — e.g. Redwood Research, the Alignment Research Center (ARC), and the UK's AI Security Institute work alongside labs to provide external safety checks.

- Governments can impose restrictions

Governance and regulation

THE INSTITUTIONAL RESPONSE

Major recent frameworks

- **EU AI Act** (2024) — risk-tiered regulation; bans on social scoring, “high-risk” categories require audit
- **US executive order 14110** (Oct 2023) — set reporting requirements for large models, but was **revoked in Jan 2025**, so US federal requirements are now in flux
- **UK AI Safety Institute** (2023, **renamed the AI Security Institute in 2025**) — pre-deployment evaluations
- **China’s interim measures** (2023) — registration + content controls

Hard open questions

- **Liability**: when an AI makes a harmful decision, who’s responsible?
- **Provenance**: how do we know what generated which content? (Cf. watermarking)
- **Concentration**: does a small number of labs running frontier models pose systemic risk?
- **Export controls**: should training compute be regulated like nuclear materials?

Labour displacement — the evidence so far

WHAT WE KNOW, WHAT WE DON'T

Studies showing displacement

- **Copywriting + translation**: Hui et al. (2023) found freelance demand **fell** after ChatGPT's release — writing jobs on Upwork down **2%**, monthly earnings down **5.2%**.
- **Illustration**: the same study finds image-related freelancers hit the same way once image models arrived.
- **Customer support**: widely reported headcount reductions through 2024–25 — trade press, not peer-reviewed evidence.

Counter-evidence + complications

- **Augmentation > replacement** in many studies (Brynjolfsson, Li & Raymond 2023).
- **Reskilling** is plausible — but the time horizon for displacement may be shorter than reskilling.
- Impact may vary more by task than industry (Acemoglu and Restrepo 2019)

It's too early to discern what's going on

Aggregate labour-market data lags by 1–3 years. Decisions are being made now – about policy, training, social safety nets – on scant/weak evidence.

Copyright, consent, provenance

- LLMs and image generators are trained on enormous corpora of human-created work, mostly **without explicit consent** of the creators
- Legal frameworks are still being written.
  - The first major AI-copyright suits arrived **~2022–2023** — the GitHub Copilot class action (Nov 2022, mostly dismissed at present), *Andersen v. Stability AI* (Jan 2023), *Getty Images v. Stability AI* (UK claim Jan 2023, US suit Feb 2023), the Authors Guild and *NYT v. OpenAI* (2023)
  - Many are still working through the courts.
- Outputs are themselves often *uncopyrightable* (in the US, at least).

> ⚠ **Provenance is increasingly a research *and* policy problem of its own.**
>
> How do we track what generated what? How do we attribute when training data is laundered through a model? How do creators opt out?

Plenary

THE SCENARIO

A research team wants to pretest policy survey questions in places where fieldwork is hard — conflict zones, closed regimes, hard-to-reach minorities. Their proposal: replace the human respondents with **synthetic respondents** — an LLM prompted to answer *as if* it were, say, a 45-year-old farmer in a named region — run across thousands of demographic profiles.

The pull

- A pilot costs pounds, not tens of thousands.
- Results arrive in an afternoon, not months.
- No enumerators at risk. No real respondents to burden.

The push

- Whose views is the model reporting? People matching the profile wrote little of the training data — the model may report what *others wrote about them*.
- Training-data bias becomes **measurement bias** — the instrument is skewed before a single question is asked.
- Nobody consented to being simulated. Who speaks for a group when a model claims to?

Note

This is live research practice, not a hypothetical — Argyle et al. (2023), on your reading list, do a version of it for US public opinion.

Plenary · what to discuss

IN YOUR GROUP — 20 MINUTES

1. What is the **quantity being measured** — public opinion, or the model's summary of text written about a group?
2. Where would you accept synthetic respondents — **pretesting question wording**, or **estimating what people believe**? Is there a line?
3. How would you **validate** the synthetic answers — and against what, if fieldwork is the thing you cannot do?
4. No respondent was contacted, so no respondent consented. Does **consent** still apply — and whose?
5. Is there a version of this project you would **refuse to take**?

Plenary · debrief

SOME THINGS PEOPLE OFTEN SURFACE IN THESE DISCUSSIONS

- **Ask what is being measured before asking whether it works.** A synthetic respondent returns a summary of text written by — and about — people fitting the profile. That is a different quantity from what those people believe.
- **The method is thinnest exactly where you want it most.** Argyle et al. (2023) get their *algorithmic fidelity* on US public opinion, where the training corpus is dense. What about in a conflict zone?
- **Between-group patterns get stamped onto individuals.** LLMs may commit ecological fallacies.
- **Inference is more than just the point estimate.** Estimating what a population believes and supplying CIs is super tricky (maybe impossible?)
- **Validating needs the fieldwork you said you couldn't do.** With no gold standard you cannot bound the bias.

Part 3 · What next + exam prep

What to read next

ONE CURATED LIST — NOT A FIREHOSE

For technical depth

- Goodfellow, Bengio & Courville — *Deep Learning* (2016). Free online. The reference text.
- Karpathy — *Neural Networks: Zero to Hero* (YouTube). The best teaching for transformers in recent memory.
- Sutton & Barto — *Reinforcement Learning* — for the RLHF half of LLMs.
- Distill.pub archive + the Hugging Face NLP course — visual and hands-on.

FOR BREADTH + SOCIETY

- Russell — *Human Compatible*.
- Crawford — *Atlas of AI*.
- Mitchell — *Artificial Intelligence: A Guide for Thinking Humans*.
- Bender, Gebru, McMillan-Major & Shmitchell — *On the Dangers of Stochastic Parrots* (Bender et al., 2021).

FOR SOCIAL SCIENCE SPECIFICALLY

- **Spirling (2023)** in *Nature* — why open-source models matter for research.
- **Argyle et al. (2023)** — simulating human respondents with LLMs.
- ⭐ **Egami et al. (2023)** — how to put LLM labels in a regression without breaking your standard errors.

Where is AI research moving?

FOR YOUR OWN NEXT-STEP PLANNING

1. **Multi-modal everything** — text, image, video, audio in one model. Already here for inputs; outputs are next.
2. ⭐ **Efficiency** — small models that match big ones. Distillation + quantisation + sparse activations.
3. **Personalisation** — models that adapt to *you* via short fine-tunes and persistent memory

Maybe we'll see *new architectures* or approaches (see LeCun's new work)?

How to *use* AI in your own work

CONCRETE HABITS WORTH FORMING

1. **Verify** every fact, citation, statistic
2. **Document your prompts** (for yourself more than anything!)
3. **Pre-register your evaluations**, and hand-code a gold-standard subset, whenever an LLM is doing measurement for you.
4. Figure out **disclosure** requirements for your context
5. **Maintain skills** over the items you delegate, you can't supervise the LLM without them

Tip

Even with Claude Fable at our fingertips, the quality of published outputs varies hugely

- This seems to be a product of *how* humans interface with the models

Quiz · the whole course in five questions

≈ 15 MIN

Pen and paper — the same equipment as Friday's exam.

1. **A forward pass.** Weights $\mathbf{w} = [1, -2]$, bias $b = 0.5$, input $\mathbf{x} = [3, 1]$. Compute $y = \mathbf{w} \cdot \mathbf{x} + b$.
2. **A parameter count.** A fully-connected layer maps 10 inputs to 4 outputs. How many parameters does it have, biases included?
3. **An output shape.** A 32×32 image goes through one convolutional layer: a 5×5 kernel, stride 1, no padding. What size is the output?
4. **Attention.** In self-attention, each token computes a query, a key, and a value. Give the role of each — one phrase per vector.
5. **Sampling.** You set the generation temperature to 0. What does the model do at each step — and what is one reason you might not want that?

Question 1 is week 1; questions 2–3 are week 2; questions 4–5 are week 3.

Quiz · answers 1–3

1 · FORWARD PASS

$$y = (1 \times 3) + (-2 \times 1) + 0.5 = 3 - 2 + 0.5 = 1.5$$

2 · PARAMETER COUNT

$$10 \times 4 \text{ weights} + 4 \text{ biases} = 40 + 4 = 44$$

3 · OUTPUT SHAPE

$$\frac{32 + 2(0) - 5}{1} + 1 = 27 + 1 = 28 \quad \Rightarrow \quad 28 \times 28$$

The same rule as lecture 6: $\lfloor (H + 2P - k)/s \rfloor + 1$.

Quiz · answers 4 and 5

4 · ATTENTION

- **Query** — *“what am I looking for?”*
- **Key** — *“what do I offer?”*
- **Value** — *“if you attend to me, here’s the content I’ll contribute.”*

5 · TEMPERATURE 0

$\tau = 0$ reduces sampling to **argmax**: at every step the model picks the single most likely token, so generation is deterministic — the same prompt gives the same text every time.

One reason not to want that: the text turns repetitive and flat, and a confident wrong answer comes back wrong on every run.

Most of this course is a dot product, a count, a shape, three vectors, and a few scalars as hyperparameters.

The final exam

FRIDAY 21 AUGUST

- **Two hours**, pen and paper. No laptops. No phones.
- **Calculators are allowed.**
- **Forty multiple-choice questions**, **six options** each, exactly one correct.
- One mark each — marked out of 40, reported as a percentage.
- **No negative marking.** A blank scores the same as a wrong answer, so **answer every question**.

Examinable

- Anything in the slides.
- Anything in the labs you completed.
- Anything in the assigned readings.

NOT EXAMINABLE

- Detailed knowledge of specific papers beyond what we covered.
- PyTorch syntax memorisation.
- The optional further-reading textbooks.

What the questions look like

TWO HOURS, FORTY QUESTIONS — ABOUT TWO MINUTES EACH IN PARTS A–B, THREE THEREAFTER

Roughly half are **calculations**

E.g. a forward pass, gradient, output shape, parameter count, softmax, cosine, three steps of BPE. You *select* the number you work out.

THE REST ARE **CODE AND JUDGEMENT**

Read a few lines of PyTorch or pseudocode and say what is wrong, or put them in the right order.

Or: here is a training curve / a validation result / a research design — what is *actually* going on?

⚠ **Arithmetic alone will not be enough.** On several questions two options carry the **same number** but differ in the claim attached to it.

- Note that **neither** of a matching pair is necessarily right – on some questions both are wrong and the answer is a third option. **Read all six options before you choose.**

The paper runs roughly in course order and **gets harder as it goes**. The last few questions are meant to stretch the strongest of you. A handful in the middle — the longer hand-traces — take noticeably more than three minutes. Ration your time and **move on and come back**.

How to study

1. Reread the **slides** from each lecture — and skim the **solution cells** of the labs you completed. Both are examinable.
2. Be able to define every term across all 12 decks.
3. Be able to **draw a computation graph** for a small expression and compute its gradient.
4. Calculate softmax + cross-entropy on paper.
5. Sleep!

A final word

A final word

THREE WEEKS AGO, AI WAS A BLACK BOX.

We've gone through *75ish years* of research progress!

- That's some workrate, so you should be pleased to have made it to the end
- Give it some time to sink in

Thank you

- To **Kaifang Zhou**, for running the labs.
- To *you*, for showing up to three hours of lecture and 90 minutes of lab for the last three weeks!

Stay in touch.

- **Send me what you build and what breaks.**

Tip

Good luck on Friday — and thank you again. It's been a privilege to teach you.

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